

A Comprehensive Survey on Ransomware Detection Using Machine Learning and Deep Learning

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Abstract

Ransomware is a type of malicious software that has been on the rise in recent years. It typically encrypts the user's files or systems and retains the decryption key until the victim pays a ransom, which is usually in cryptocurrency due to its untraceability. Ransomware attacks are increasing in tandem with the internet's rapid growth. Researchers and businesses are focusing on various methods for detecting and preventing ransomware attacks. As advanced ransomware was created using AI-based technology, a new intelligence method should be implemented to prevent such attacks. Static and dynamic analysis techniques are used to examine ransomware. The efficiency with which sample features are extracted determines the analysis. Malware researchers are concentrating their efforts on deep learning and machine learning methods for detecting ransomware. In this paper, we discussed deep learning and machine learning methods, as well as various algorithms and challenges for detecting ransomware.

Keywords: threat, bitcoin, machine learning, deep learning,

1. Introduction

The first step in detecting ransomware is to use automation and malware analysis to find dangerous files early in the kill chain. The cyber kill chain is a cybersecurity concept that monitors the stages of a cyber-attack, finds weaknesses, and assists security teams in stopping attacks at each level. The detection and analysis procedure normally includes compiling a list of potentially suspicious behaviors in order to determine whether or not something malicious is taking place.

Deep learning and machine learning have an impact on all aspects of life. Because of their decision-making ability, these technologies have been used in every domain. These have a wide range of applications in computing and are becoming increasingly important in cybersecurity. Because of the use of these technologies, advanced attacks and threat detection have become easier and faster. Both were effective in pattern recognition. Machine learning and deep learning are used to detect a variety of malware and ransomware.

1.1. Ransomware detection Approach:

- 1.1.1. **Signature-Based Detection:** This is a technique that many antivirus solutions use. A signature is a piece of code that allows a specific strain of ransomware to be identified. The pattern enables the software to detect and stop an attack in real-time. It is useful for detecting old ransomware strains, but it does not protect against new ransomware strains.
- 1.1.2. **Behavior-Based Detection:** It monitors file executions for abnormalities using behavior-based detection. It runs the file and watches for malicious behavior such as overwriting DLL files, encrypting algorithms, or creating malicious links. The main methods for monitoring data behavior are file system changes, traffic analysis, and API call examination. The main disadvantage of this method is that it takes time to analyze data behavior.
- 1.1.3. **Deception-Based Detection:** This method relies on tricking adversaries to detect ransomware by setting up honeypots. It will serve as a trap for attackers. [16] The study focused on locky ransomware. Using the process monitoring technique, decoy files were placed in each directory. A recursive and depth-first search was used to traverse the file. When an attacker activates the decoy file, malicious activity occurs.

2. Related work

This study is about ransomware detection utilizing machine learning and deep learning to provide security insights that can help end users detect ransomware before it encrypts and exfiltrates crucial data.

CF-NCF (Class Frequency - Non-Class Frequency) was proposed by Seong II Bae et al [1], feature vectors were created using CF-NCF, and six machine learning techniques were tested to detect ransomware. CF-NCF is a classification model indicator that is based on TF-IDF (Term Frequency - Inverse Document Frequency). The suggested ransomware detection approach can distinguish ransomware from both benign and malicious files. It was published in 2019 and used 300 benign files, 1000 ransomware samples, and 900 malware samples as sample input files. The proposed method can detect ransomware n files with an accuracy of up to 98.65%.

S.H. Kok et al. 2020 [2] present a pre-encryption detection technique for evaluating and detecting ransomware before any encryption occurs. There are two detection levels in the pre-encryption detection algorithm. The first is a signature repository that detects any signature matches with existing malware. The second stage of detection employs a learning algorithm capable of detecting both known and undiscovered ransomware. As datasets, 491 ransomware samples (Old) and 942 goodware samples were employed. Additional ransomware samples were obtained from online sources, namely VirusTotal and theZoo, for a total of 357 ransomware samples from VirusTotal and 56 ransomware samples from theZoo. To train the model, the learning method employs a machine learning approach based on data from the application programming interface. To collect all of the application program interface requests in each sample, the retrieved samples were dynamically analyzed using the Cuckoo Sandbox analysis system. This data was retrieved and transformed into a dataset format for machine learning, which was then used to train the module and test the learning method from the pre-encryption detection technique. Based on the results of the experiment, it is possible to infer that the learning algorithm achieved its goal of detecting ransomware prior to encryption and that its performance is reliable with a high net benefit.

Greg Cusack et al.[3] suggest a solution that takes advantage of a current innovation in networking technology, programmable forwarding engines (PFEs), which enable the collection of per-packet network monitoring data at high speeds. They collect data in order to monitor network activity between infected computers and command and control (C&C) servers. They collect information from traffic and utilize the data to classify malware. Over 100MB of ransomware traffic traces were gathered from malware traffic-analysis.net for the dataset, resulting in 265 distinct bidirectional ransomware-related flows. They collect another 100MB of malware-free network traffic to use as a baseline. These statistics are made up of flows related to online surfing, file streaming, and file downloading. This model has a detection rate of more than 0.86 and a false negative rate of less than 0.11. As a result, the

findings indicate that a flow-based fingerprinting method is more viable and accurate for detecting ransomware before encryption.

Sonu B et al. [8] focus on ransomware attacks on Windows, Android, and other platforms. In Windows OS and Android, ransomware attackers can be discovered by monitoring abnormal file systems and paying particular attention to the android manifest file. Some security solutions have been developed to combat ransomware and to improve cyber security, and they operate in four stages. Understanding the risk, recognizing the attack surface, and identifying weak areas are all part of the predicate. Preventing incidents entails limiting the attack surface and preventing incidents. Responding involves reacting to breaches and mitigating damage, analyzing and learning, and detecting involves spotting incidents and threats and isolating and controlling them.

Most malware researchers attempt to detect ransomware by observing the program's behavior during execution. As a result, these approaches are costly and necessitate the expenditure of resources to execute each program. FarnoushManavi et al. [14] describe ransomware detection without launching the software or performing any special pre-processing, instead of relying solely on the executable file's headers. The suggested method generates a graph from the headers of portable executable files, which is then mapped in an eigenspace using the "Power Iteration" method. This mapping transforms a portable executable file into a feature vector that is then used to train a Random Forest classifier. As compared to previous approaches, the key advantages of the proposed system include acceptable processing complexity in big datasets and high detection rates.

Hong-Yi Chang et al. [10] propose a ransomware prediction system based on the weighted K Nearest-Neighbor algorithm, which incorporates ransomware packet traffic identification and prediction, as well as the design and implementation of a dynamic isolation system connected with Software-Defined Networking. The results demonstrate that the detection precision for normal and abnormal flow is 99.7 and 97.7, respectively.

T.R.Reshmi [15] gives a systematic review of ransomware-related information security breaches. She highlighted many varieties of ransomware as well as the encryption mechanism used by each ransomware, infection, attack techniques, impacts, possible mitigation, and prevention from the assaults, which were primarily focused on the Windows operating system

Mohammad Mehdi Ahmadian et al. [17] suggested a new method for preventing and detecting highly survivable ransomware in backup systems. This system was eligible for data recovery after it was compromised by a ransomware attack. To detect ransomware, the entropy of six different file types was determined. Machine learning and deep learning were used to detect and classify ransomware threats based on the ideal entropy reference value. The entropy of six different file formats was tested using 600 normal files and 600 attack-

infected ones. The technology was also capable of recouping losses caused by a compromised attack. This method also allows for the restoration of original data files in the event of a ransomware attack.

[18] provided a comprehensive behavioral study of crypto-ransomware networks. Locky ransomware was employed to examine the behavioral analysis and impacts. This study classified network flows and network packets using four machine learning classifiers: Random Forest, Library for Support Vector Machines, random tree, and Bayes Net. The dedicated sandbox was used to examine network behaviors. Network packets were captured in order to create PCAP files, which were then utilized to extract network features. The suggested detection method has a high detection accuracy, a low false-positive rate, and is extremely effective in tracking ransomware network activity.

3. Commonly Used Algorithm

Some of the commonly used machine learning algorithms for training the classifier are as follows.

- 3.1. **K-Nearest Neighbour (K-NN):**KNN is a supervised machine learning technique that is used to address regression and classification issues. It classifies the input instance by taking into account the class label of the k closest training instances. The class of majority instances is predicted to be the class of input instances.
- 3.2. **Logistic Regression:**Logistic Regression is a Machine Learning technique for solving classification issues; it is a predictive analytic approach based on the idea of probability. It forecasts categorical dependent variables based on a collection of independent variables.
- 3.3. **Artificial Neural Networks (ANNs):**ANNs are nonlinear statistical models that show a relationship between inputs and outputs in order to uncover a new pattern. It is mostly employed in the approximation of random functions.
- 3.4. **Decision Tree:**A Decision Tree is a supervised learning approach used to address Classification issues.It is a tree-structured classifier, with each leaf node representing the outcome, branches representing the decision rules, and interior nodes representing dataset attributes. A tree is built using the Classification and Regression Tree (CART) technique.
- 3.5. **Random Forest:**Random Forest is a machine learning ensemble bagging method that belongs to the supervised learning technique and is used for both classification and regression issues. It consists of a number of decision trees on distinct subsets of the supplied dataset and takes the average to increase the dataset's predicting accuracy.

- 3.6. **Boosting Algorithms:** Boosting is another form of ensemble machine learning technique that sequentially trains numerous weak classifiers. The weak classifiers are trained without replacement in linear order. After that, the average of all weak classifiers is utilized to generate a strong classifier. Gradient Boosting (GB) and Adaptive Boosting (AB) are two popular boosting techniques (AdaBoost).

4. Challenges to implementing DL in Ransomware Detection

Ransomware detection and prevention has recently become a hot topic. Deep learning approaches are faster and more effective at predicting ransomware. However, there hasn't been a lot of research done on ransomware detection using deep learning in various industries. Because it almost has an impact on the digital world. The main problem with detecting ransomware is the extraction of features from PE files for deep learning and machine learning.

Another issue with implementing ransomware detection was the high computational cost of updating and modifying the system on a regular basis. Because the malware was created every hour and its behavior was changed, it is complicated and costly when compared to other detection techniques.

Nowadays, ransomware is created using machine learning techniques to avoid detection. Developers can avoid the machine-learning detection system discussed by Kolosnjaji et al. [11] by employing an evasion attack.

Predicting ransomware evolution will be a challenge that will be critical in the future when detecting ransomware. The incorporation of AI into ransomware attacks poses the most difficult challenge of all. With attackers employing similar AI techniques to those used by AI-based defense systems, defensive measures must be configured to account for adaptable AI-based attack vectors.

5. Conclusion

In this paper, we presented a detailed review of ransomware and related topics, as well as ransomware detection methodologies based on machine learning and deep learning technology. The combination of machine learning and deep learning technologies results in a robust and accurate ransomware detection solution. Feature extraction and classification are the two most important processes in ransomware detection. With enough updated training data, the model should be able to be created. This study provides a user manual that malware researchers can use as a reference to work with available technology in the field of ransomware detection. It can help them design more efficient ransomware detection algorithms as they are considering the potential remedies.

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